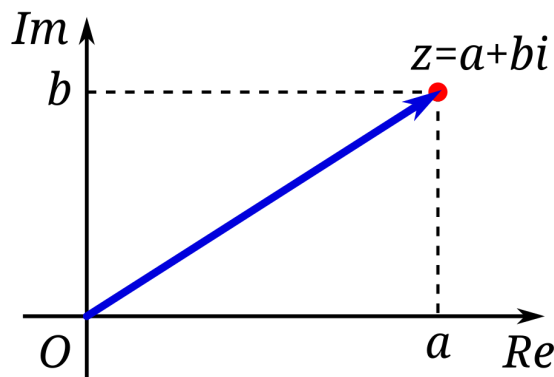


Комплексное число

В математике **комплексное число** — это элемент **системы счисления**, которая расширяет **действительные числа** за счёт особого элемента, обозначаемого i , называемого **мнимой единицей** и удовлетворяющего уравнению $i^2 = -1$. Поскольку ни одно действительное число не удовлетворяет этому уравнению, i было названо **мнимым числом Рене Декартом**. Любое комплексное число можно представить в виде $a + bi$, где a и b — действительные числа, a называется **действительная часть**, а b называется **мнимая часть**. Множество комплексных чисел обозначается символами $\mathbb{C}$ или $\mathbf{C}$. Несмотря на историческую терминологию,^[1] «мнимые» числа не являются вымышленными: с математической точки зрения они не менее реальны, чем действительные числа,^{[2][3]} и они необходимы для научного описания физического мира.^[4]



Комплексное число Z можно наглядно представить в виде пары чисел (a, b) , образующих **вектор положения** (синий) или точку (красная) на диаграмме, называемой диаграммой Арганда, которая представляет комплексную плоскость. Re — это действительная ось, Im — мнимая ось, а i — «мнимая единица», которая удовлетворяет условию $i^2 = -1$.

Комплексные числа позволяют найти решения для всех **полиномиальных уравнений**, даже для тех, которые не имеют решений в действительных числах. Точнее, **основная теорема алгебры** утверждает, что любое непостоянное полиномиальное уравнение с действительными или комплексными коэффициентами имеет решение, являющееся комплексным числом. Например, уравнение $(x + 1)^2 = -9$ не имеет действительных решений, поскольку квадрат действительного числа не может быть отрицательным, но имеет два комплексных решения, не являющихся действительными: $x = -1 + 3i$ и $x = -1 - 3i$.

Сложение, вычитание и умножение комплексных чисел определяются с помощью правила $i^2 = -1$, а также **ассоциативного**, **коммутативного** и **дистрибутивного законов**.

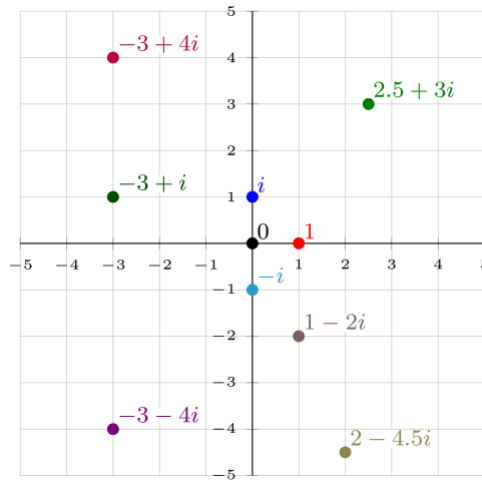
У каждого ненулевого комплексного числа есть **множитель, обратный при умножении**, что позволяет делить комплексные числа, кроме нуля, на другие комплексные числа. Это делает комплексные числа **полем**, в котором вещественные числа являются подмножеством. Благодаря этим свойствам $a + bi = a + ib$, и то, какая форма используется, зависит от традиций и стиля.

Комплексные числа также образуют **вещественное векторное пространство размерности два**, в котором $\{1, i\}$ является **стандартным базисом**. Этот стандартный базис превращает комплексные числа в **декартову плоскость**, называемую **комплексной плоскостью**. Это позволяет геометрически интерпретировать комплексные числа и операции над ними, и наоборот, некоторые геометрические объекты и операции можно выразить через комплексные числа. Например, действительные числа образуют **действительную прямую**, которая изображается в виде горизонтальной оси комплексной плоскости, а действительные кратные i — в виде вертикальной оси. Комплексное число также можно определить с помощью его геометрических **полярных координат**: радиус называется **абсолютным значением** комплексного числа, а угол между положительной действительной осью и числом — **аргументом** комплексного числа. Комплексные числа с абсолютным значением, равным единице, образуют **единичный круг**. Добавление фиксированного комплексного числа ко всем комплексным числам определяет **сдвиг** в комплексной плоскости, а умножение на фиксированное комплексное число — **подобие** с центром в начале координат (размножение на абсолютное значение и поворот на аргумент). Операция **комплексного сопряжения** является **симметрией отражения** относительно действительной оси.

Комплексные числа образуют сложную структуру, которая одновременно является **алгебраически замкнутым полем**, **коммутативной алгеброй** над вещественными числами и **евклидовым векторным пространством** размерности два.



Определение и основные операции



Various complex numbers depicted in the complex plane.



A complex number is an expression of the form $a + bi$, where a and b are real numbers, and i is an abstract symbol, the so-called imaginary unit, whose meaning will be explained further below.^[5] For example, $2 + 3i$ is a complex number.^[6]

For a complex number $a + bi$, the real number a is called its *real part*, and the real number b (not the complex number bi) is its *imaginary part*.^{[7][8]} The real part of a complex number z is denoted $\text{Re}(z)$, $\mathcal{R}e(z)$, or $\Re(z)$; the imaginary part is $\text{Im}(z)$, $\mathcal{I}m(z)$, or $\Im(z)$: for example, $\mathbf{Re}(2 + 3i) = 2$, $\mathbf{Im}(2 + 3i) = 3$.

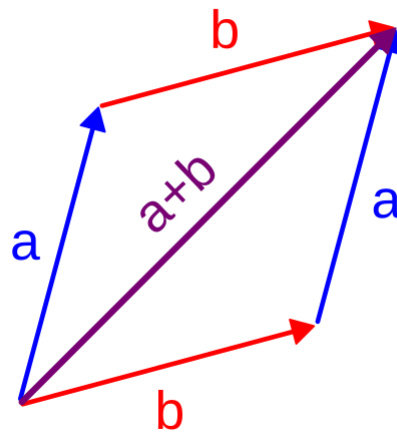
A complex number z can be identified with the **ordered pair** of real numbers $(\Re(z), \Im(z))$, which may be interpreted as coordinates of a point in a Euclidean plane with standard coordinates, which is then called the **complex plane** or **Argand diagram**.^{[9][10][a]} The horizontal axis is generally used to display the real part, with increasing values to the right, and the imaginary part marks the vertical axis, with increasing values upwards.

A real number a can be regarded as a complex number $a + 0i$, whose imaginary part is 0. A purely imaginary number bi is a complex number $0 + bi$, whose real part is zero. It is common to write $a + 0i = a$, $0 + bi = bi$, and $a + (-b)i = a - bi$; for example, $3 + (-4)i = 3 - 4i$.

The **set** of all complex numbers is denoted by $\mathbb{C}$ (**blackboard bold**) or $\mathbf{C}$ (**upright bold**).

In some disciplines such as electromagnetism and electrical engineering, j is used instead of i , as i frequently represents electric current,^{[11][12]} and complex numbers are written as $a + bj$ or $a + jb$.

Addition and subtraction



Addition of two complex numbers can be done geometrically by constructing a parallelogram.

Two complex numbers $a = x + yi$ and $b = u + vi$ are added by separately adding their real and imaginary parts.^[13] That is to say:



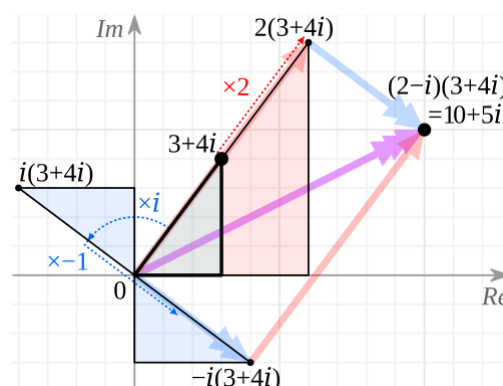
$$a + b = (x + yi) + (u + vi) = (x + u) + (y + v)i.$$

Similarly, subtraction can be performed as

$$a - b = (x + yi) - (u + vi) = (x - u) + (y - v)i.$$

The addition can be geometrically visualized as follows: the sum of two complex numbers a and b , interpreted as points in the complex plane, is the point obtained by building a parallelogram from the three vertices O , and the points of the arrows labeled a and b (provided that they are not on a line). Equivalently, calling these points A, B , respectively and the fourth point of the parallelogram X the triangles OAB and XBA are congruent.

Multiplication



Multiplication of complex numbers $2-i$ and $3+4i$ visualized with vectors

The product of two complex numbers is computed as follows:^[13]

$$(a + bi) \cdot (c + di) = ac - bd + (ad + bc)i.$$

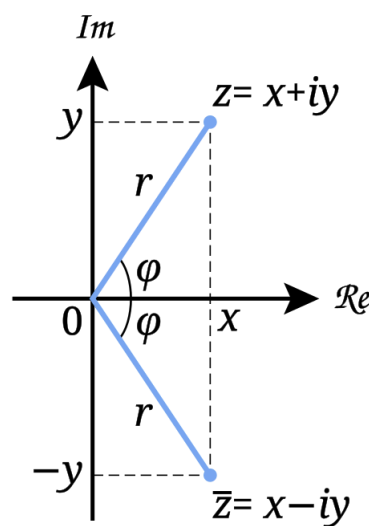
For example, $(2 - i)(3 + 4i) = 2 \cdot 3 - ((-1) \cdot 4) + (2 \cdot 4 + (-1) \cdot 3)i = 10 + 5i$. In particular, this includes as a special case the fundamental formula

$$i^2 = i \cdot i = -1.$$

This formula distinguishes the complex number i from any real number, since the square of any (negative or positive) real number is always a non-negative real number.

With this definition of multiplication and addition, familiar rules for the arithmetic of rational or real numbers continue to hold for complex numbers. More precisely, the [distributive property](#), the [commutative properties](#) (of addition and multiplication) hold. Therefore, the complex numbers form an algebraic structure known as a [field](#), the same way as the rational or real numbers do.^[14]

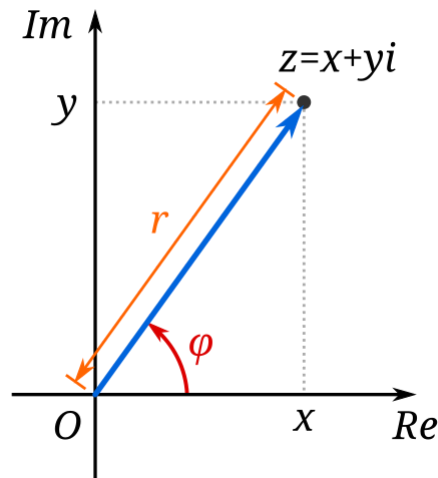
Complex conjugate, absolute value, argument and division



Geometric representation of z and its conjugate $\bar{z}$ in the complex plane.

The [complex conjugate](#) of the complex number $z = x + yi$ is defined as $\bar{z} = x - yi$.^[15] It is also denoted by some authors by z^* . Geometrically, $\bar{z}$ is the "reflection" of z about the real axis.

Conjugating twice gives the original complex number: $\overline{\bar{z}} = z$. A complex number is real [if and only if](#) it equals its own conjugate. The [unary operation](#) of taking the complex conjugate of a complex number cannot be expressed by applying only the basic operations of addition, subtraction, multiplication and division.



Argument φ and modulus r locate a point in the complex plane.

For any complex number $Z = x + yi$, the product

$$z \cdot \bar{z} = (x + iy)(x - iy) = x^2 + y^2$$

is a *non-negative real* number. This allows to define the *absolute value* (or *modulus* or *magnitude*) of z to be the square root^[16]

$$|z| = \sqrt{x^2 + y^2}.$$

By *Pythagoras' theorem*, $|z|$ is the distance from the origin to the point representing the complex number z in the complex plane. In particular, the *circle of radius one* around the origin consists precisely of the numbers z such that $|z| = 1$, known as the **unit complex numbers**. If $z = x = x + 0i$ is a real number, then $|z| = |x|$: its absolute value as a complex number and as a real number are equal.

Using the conjugate, the *reciprocal* of a nonzero complex number $z = x + yi$ can be computed to be^[13]

$$\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{x - yi}{x^2 + y^2} = \frac{x}{x^2 + y^2} - \frac{y}{x^2 + y^2}i.$$

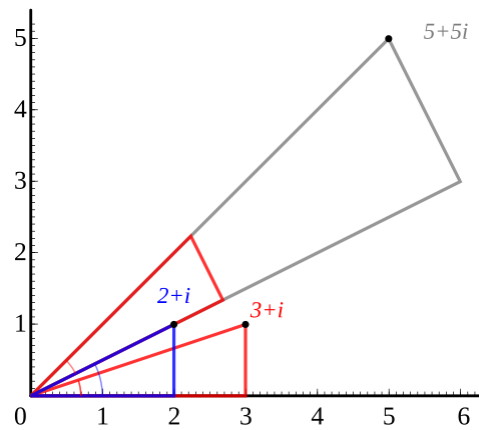
More generally, the division of an arbitrary complex number $w = u + vi$ by a non-zero complex number $z = x + yi$ equals

$$\frac{w}{z} = \frac{w\bar{z}}{|z|^2} = \frac{(u + vi)(x - iy)}{x^2 + y^2} = \frac{ux + vy}{x^2 + y^2} + \frac{vx - uy}{x^2 + y^2}i.$$

This process is sometimes called "*rationalization*" of the denominator (although the denominator in the final expression may be an irrational real number), because it resembles the method to remove roots from simple expressions in a denominator.^{[17][18]}

The **argument** of Z (sometimes called the "phase" φ)^[10] is the angle of the **radius** OZ with the positive real axis, and is written as $\arg Z$, expressed in **radians** in this article. The angle is defined only up to adding integer multiples of 2π , since a rotation by 2π (or 360°) around the origin leaves all points in the complex plane unchanged. One possible choice to uniquely specify the argument is to require it to be within the interval $(-\pi, \pi]$, which is referred to as the **principal value**.^[19] The argument can be computed from the rectangular form $x + yi$ by means of the **arctan** (inverse tangent) function.^[20]

Polar form



Multiplication of $2 + i$ (blue triangle) and $3 + i$ (red triangle). The red triangle is rotated to match the vertex of the blue one (the adding of both angles in the terms $\varphi_1 + \varphi_2$ in the equation) and stretched by the length of the **hypotenuse** of the blue triangle (the multiplication of both radiuses, as per term $r_1 r_2$ in the equation).

For any complex number z , with absolute value $r = |z|$ and argument φ , the equation^[13]

$$z = r(\cos \varphi + i \sin \varphi)$$

holds. This identity is referred to as the polar form of z . It is sometimes abbreviated as $z = r \text{cis } \varphi$. In electronics, one represents a **phasor** with amplitude r and phase φ in **angle notation**.^[21]

$$z = r \angle \varphi.$$

If two complex numbers are given in polar form, i.e., $z_1 = r_1(\cos \varphi_1 + i \sin \varphi_1)$ and $z_2 = r_2(\cos \varphi_2 + i \sin \varphi_2)$, the product and division can be computed as

$$z_1 z_2 = r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2)).$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2)), \text{ if } z_2 \neq 0.$$

(These are a consequence of the **trigonometric identities** for the sine and cosine function.) In other words, the absolute values are *multiplied* and the arguments are *added* to yield the polar

form of the product. The picture at the right illustrates the multiplication of

$$(2 + i)(3 + i) = 5 + 5i.$$

Because the real and imaginary part of $5 + 5i$ are equal, the argument of that number is 45 degrees, or $\pi/4$ (in [radian](#)). On the other hand, it is also the sum of the angles at the origin of the red and blue triangles are [arctan](#)(1/3) and [arctan](#)(1/2), respectively. Thus, the formula

$$\frac{\pi}{4} = \arctan\left(\frac{1}{2}\right) + \arctan\left(\frac{1}{3}\right)$$

holds. As the [arctan](#) function can be approximated highly efficiently, formulas like this – known as [Machin-like formulas](#) – are used for high-precision approximations of π .^[22]

$$\frac{\pi}{4} = 4 \arctan\left(\frac{1}{5}\right) - \arctan\left(\frac{1}{239}\right)$$

Powers and roots

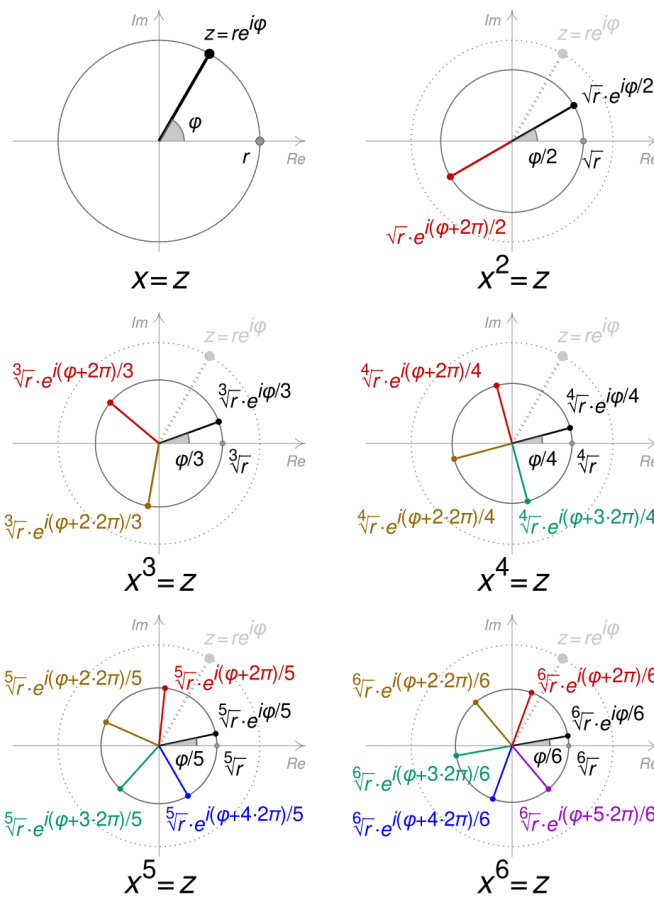
The n -th power of a complex number can be computed using [de Moivre's formula](#), which is obtained by repeatedly applying the above formula for the product:^[23]

$$z^n = \underbrace{z \cdot \dots \cdot z}_{n \text{ factors}} = (r(\cos \varphi + i \sin \varphi))^n = r^n (\cos n\varphi + i \sin n\varphi).$$

For example, the first few powers of the imaginary unit i are

$$i, i^2 = -1, i^3 = -i, i^4 = 1, i^5 = i, \dots$$





Geometric representation of the 2nd to 6th roots of a complex number Z , in polar form $re^{i\phi}$ where $r = |Z|$ and $\phi = \arg Z$. If Z is real, $\phi = 0$ or π . Principal roots are shown in black.

The n **n th roots** of a complex number Z are given by^[24]

$$z^{1/n} = \sqrt[n]{r} \left(\cos \left(\frac{\varphi + 2k\pi}{n} \right) + i \sin \left(\frac{\varphi + 2k\pi}{n} \right) \right)$$

for $0 \leq k \leq n - 1$. (Here $\sqrt[n]{r}$ is the usual (positive) n th root of the positive real number r .)

Because sine and cosine are periodic, other integer values of k do not give other values. For any $z \neq 0$, there are, in particular n distinct complex n -th roots. For example, there are 4 fourth roots of 1, namely

$$z_1 = 1, z_2 = i, z_3 = -1, z_4 = -i.$$

In general there is *no* natural way of distinguishing one particular complex n th root of a complex number. (This is in contrast to the roots of a positive real number x , which has a unique positive real n -th root, which is therefore commonly referred to as *the* n -th root of x .) One refers to this situation by saying that the n th root is an **n -valued function** of Z .

Fundamental theorem of algebra

The **fundamental theorem of algebra**, of **Carl Friedrich Gauss** and **Jean le Rond d'Alembert**, states that for any complex numbers (called **coefficients**) $a_0, \dots, a_n$, the equation

$$a_n z^n + \dots + a_1 z + a_0 = 0$$

has at least one complex solution z , provided that at least one of the higher coefficients $a_1, \dots, a_n$ is nonzero.^[25] This property does not hold for the [field of rational numbers](#) $\mathbb{Q}$ (the polynomial $x^2 - 2$ does not have a rational root, because $\sqrt{2}$ is not a rational number) nor the real numbers $\mathbb{R}$ (the polynomial $x^2 + 4$ does not have a real root, because the square of x is positive for any real number x).

Because of this fact, $\mathbb{C}$ is called an [algebraically closed field](#). It is a cornerstone of various applications of complex numbers, as is detailed further below. There are various proofs of this theorem, by either analytic methods such as [Liouville's theorem](#), or [topological](#) ones such as the [winding number](#), or a proof combining [Galois theory](#) and the fact that any real polynomial of odd degree has at least one real root.

The field of complex numbers is defined as the (unique) algebraic [extension field](#) of the real numbers later in [#Abstract algebraic definitions](#).



История

The solution in [radicals](#) (without [trigonometric functions](#)) of a general [cubic equation](#), when all three of its roots are real numbers, contains the square roots of [negative numbers](#), a situation that cannot be rectified by factoring aided by the [rational root test](#), if the cubic is [irreducible](#); this is the so-called [casus irreducibilis](#) ('irreducible case'). This conundrum led Italian mathematician [Gerolamo Cardano](#) to conceive of complex numbers in around 1545 in his [Ars Magna](#),^[26] though his understanding was rudimentary; moreover, he later described complex numbers as being "as subtle as they are useless".^[27] Cardano did use imaginary numbers, but described using them as "mental torture".^[28] This was prior to the use of the graphical complex plane. Cardano and other Italian mathematicians, notably [Scipione del Ferro](#), in the 1500s created an algorithm for solving cubic equations which generally had one real solution and two solutions containing an imaginary number. Because they ignored the answers with the imaginary numbers, Cardano found them useless.^[29]

Work on the problem of general polynomials ultimately led to the fundamental theorem of algebra, which shows that with complex numbers, a solution exists to every [polynomial equation](#) of degree one or higher. Complex numbers thus form an [algebraically closed field](#), where any polynomial equation has a [root](#).

Many mathematicians contributed to the development of complex numbers. The rules for addition, subtraction, multiplication, and root extraction of complex numbers were developed by the Italian mathematician [Rafael Bombelli](#).^[30] A more abstract formalism for the complex numbers was further developed by the Irish mathematician [William Rowan Hamilton](#), who extended this abstraction to the theory of [quaternions](#).^[31]

The earliest fleeting reference to [square roots](#) of [negative numbers](#) can perhaps be said to occur in the work of the Greek mathematician [Hero of Alexandria](#) in the 1st century AD, where in his [Stereometrica](#) he considered, apparently in error, the volume of an impossible [frustum](#) of a [pyramid](#) to arrive at the term $\sqrt{81 - 144}$ in his calculations, which today would simplify to $\sqrt{-63} = 3i\sqrt{7}$.^[b] Negative quantities were not conceived of in [Hellenistic mathematics](#) and Hero merely replaced the negative value by its positive $\sqrt{144 - 81} = 3\sqrt{7}$.^[33]

The impetus to study complex numbers as a topic in itself first arose in the 16th century when [algebraic solutions](#) for the roots of [cubic](#) and [quartic polynomials](#) were discovered by Italian mathematicians ([Niccolò Fontana Tartaglia](#) and [Gerolamo Cardano](#)). It was soon realized (but proved much later)^[34] that these formulas, even if one were interested only in real solutions, sometimes required the manipulation of square roots of negative numbers. In fact, it was proved later that the use of complex numbers is [unavoidable](#) when all three roots are real and distinct.^[c] However, the general formula can still be used in this case, with some care to deal with the ambiguity resulting from the existence of three cubic roots for nonzero complex numbers. Rafael Bombelli was the first to address explicitly these seemingly paradoxical solutions of cubic equations and developed the rules for complex arithmetic, trying to resolve these issues.

The term "imaginary" for these quantities was coined by [René Descartes](#) in 1637, who was at pains to stress their unreal nature.^[35]

... sometimes only imaginary, that is one can imagine as many as I said in each equation, but sometimes there exists no quantity that matches that which we imagine.

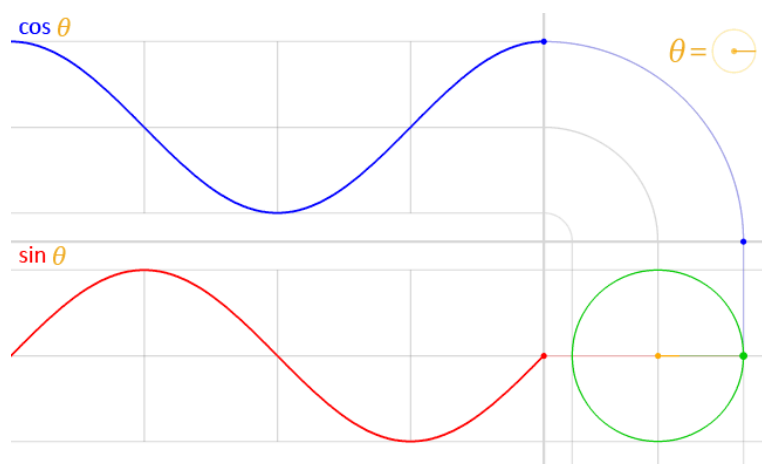
[... quelquefois seulement imaginaires c'est-à-dire que l'on peut toujours en imaginer autant que j'ai dit en chaque équation, mais qu'il n'y a quelquefois aucune quantité qui corresponde à celle qu'on imagine.]

A further source of confusion was that the equation $\sqrt{-1}^2 = \sqrt{-1}\sqrt{-1} = -1$ seemed to be capriciously inconsistent with the algebraic identity $\sqrt{a}\sqrt{b} = \sqrt{ab}$, which is valid for non-negative real numbers a and b , and which was also used in complex number calculations with one of a, b positive and the other negative. The incorrect use of this identity in the case when both a and b are negative, and the related identity $\frac{1}{\sqrt{a}} = \sqrt{\frac{1}{a}}$, even bedeviled [Leonhard Euler](#).

This difficulty eventually led to the convention of using the special symbol i in place of $\sqrt{-1}$ to guard against this mistake.^{[36][37]} Even so, Euler considered it natural to introduce students to complex numbers much earlier than we do today. In his elementary algebra text book, [Elements of Algebra](#), he introduces these numbers almost at once and then uses them in a natural way throughout.

In the 18th century complex numbers gained wider use, as it was noticed that formal manipulation of complex expressions could be used to simplify calculations involving trigonometric functions. For instance, in 1730 [Abraham de Moivre](#) noted that the identities relating trigonometric functions of an integer multiple of an angle to powers of trigonometric functions of that angle could be re-expressed by the following [de Moivre's formula](#):

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$



Euler's formula relates the complex exponential function of an imaginary argument, which can be thought of as describing [uniform circular motion](#) in the complex plane, to the cosine and sine functions, geometrically its projections onto the real and imaginary axes, respectively.

In 1748, Euler went further and obtained [Euler's formula](#) of [complex analysis](#):^[38]

$$e^{i\theta} = \cos \theta + i \sin \theta$$

by formally manipulating complex [power series](#) and observed that this formula could be used to reduce any trigonometric identity to much simpler exponential identities.

The idea of a complex number as a point in the complex plane was first described by [Danish–Norwegian](#) mathematician [Caspar Wessel](#) in 1799,^[39] although it had been anticipated as early as 1685 in [Wallis's A Treatise of Algebra](#).^[40]

Wessel's memoir appeared in the Proceedings of the [Copenhagen Academy](#) but went largely unnoticed. In 1806 [Jean-Robert Argand](#) independently issued a pamphlet on complex numbers and provided a rigorous proof of the [fundamental theorem of algebra](#).^[41] [Carl Friedrich Gauss](#) had earlier published an essentially [topological](#) proof of the theorem in 1797 but expressed his doubts at the time about "the true metaphysics of the square root of -1 ".^[42] It was not until 1831 that he overcame these doubts and published his treatise on complex numbers as points in the plane,^[43] largely establishing modern notation and terminology.^[44]

If one formerly contemplated this subject from a false point of view and therefore found a mysterious darkness, this is in large part attributable to clumsy terminology. Had one not called $+1$, -1 , $\sqrt{-1}$ positive, negative, or imaginary (or even impossible) units, but instead, say, direct, inverse, or lateral units, then there could scarcely have been talk of such darkness.

In the beginning of the 19th century, other mathematicians discovered independently the geometrical representation of the complex numbers: Buée,^{[45][46]} Mourey,^[47] Warren,^{[48][49][50]} Français and his brother, Bellavitis.^{[51][52]}

The English mathematician G.H. Hardy remarked that Gauss was the first mathematician to use complex numbers in "a really confident and scientific way" although mathematicians such as Norwegian Niels Henrik Abel and Carl Gustav Jacob Jacobi were necessarily using them routinely before Gauss published his 1831 treatise.^[53]



Augustin-Louis Cauchy and Bernhard Riemann together brought the fundamental ideas of complex analysis to a high state of completion, commencing around 1825 in Cauchy's case.

The common terms used in the theory are chiefly due to the founders. Argand called $\cos \varphi + i \sin \varphi$ the *direction factor*, and $r = \sqrt{a^2 + b^2}$ the *modulus*,^{[d][54]} Cauchy (1821) called $\cos \varphi + i \sin \varphi$ the *reduced form* (l'expression réduite)^[55] and apparently introduced the term *argument*; Gauss used i for $\sqrt{-1}$,^[e] introduced the term *complex number* for $a + bi$,^[f] and called $a^2 + b^2$ the *norm*.^[g] The expression *direction coefficient*, often used for $\cos \varphi + i \sin \varphi$, is due to Hankel (1867),^[59] and *absolute value*, for *modulus*, is due to Weierstrass.

Later classical writers on the general theory include Richard Dedekind, Otto Hölder, Felix Klein, Henri Poincaré, Hermann Schwarz, Karl Weierstrass and many others. Important work (including a systematization) in complex multivariate calculus has been started at beginning of the 20th century. Important results have been achieved by Wilhelm Wirtinger in 1927.

Абстрактные и алгебраические определения

While the above concrete definitions, including the addition and multiplication, accurately describe the complex numbers, there are other, equivalent approaches that reveal the abstract algebraic structure of the complex numbers more immediately.

One definition of the complex numbers is that they form a field denoted $\mathbb{C}$ that contains the real field $\mathbb{R}$, and is generated over $\mathbb{R}$ by a distinguished element denoted i such that $i^2 = -1$. Equivalently, $\mathbb{C}$ is the *splitting field* of the polynomial $x^2 + 1$ over the real field $\mathbb{R}$.^{[60][61][62]}

Existence of the field can be established by means of various models. One model is to regard the complex field as the set $\mathbb{R}^2$ equipped with addition and multiplication laws on ordered pairs (a, b) and (a', b') :

$$(a, b) + (a', b') = (a + a', b + b'), \quad (a, b)(a', b') = (aa' - bb', ab' + a'b).$$

In this model the element i corresponds to the element $(0, 1)$, and the real numbers are embedded as the set of pairs $(a, 0)$. Another is the standard construction of a splitting field using a [quotient](#) of the [polynomial ring](#) $\mathbb{R}[X]$, given below.

Because the field $\mathbb{C}$ has different models, they are not all literally the same mathematical object, but they are all [isomorphic](#), up to a [ring isomorphism](#) preserving the real numbers. More precisely, given two models $\mathbb{C}_1 = \mathbb{R}(i_1)$ and $\mathbb{C}_2 = \mathbb{R}(i_2)$, there is a unique isomorphism $\phi : \mathbb{C}_1 \rightarrow \mathbb{C}_2$ which is linear over the real numbers and such that $\phi(i_1) = i_2$.^[63] More generally, any non-trivial finite extension field of the reals is isomorphic to the complex field.^{[64][65]} Moreover, the isomorphism is unique, provided it preserves the real subfield, up to the action of the [Galois group](#) (complex conjugation).



Construction as a quotient ring

One approach to $\mathbb{C}$ is via [polynomials](#), i.e., expressions of the form

$$p(X) = a_n X^n + \cdots + a_1 X + a_0,$$

where the [coefficients](#) $a_0, \dots, a_n$ are real numbers. The set of all such polynomials is denoted by $\mathbb{R}[X]$. Since sums and products of polynomials are again polynomials, this set $\mathbb{R}[X]$ forms a [commutative ring](#), called the [polynomial ring](#) (over the reals). To every such polynomial p , one may assign the complex number $p(i) = a_n i^n + \cdots + a_1 i + a_0$, i.e., the value obtained by setting $X = i$. This defines a function

$$\mathbb{R}[X] \rightarrow \mathbb{C}$$

This function is [surjective](#) since every complex number can be obtained in such a way: the evaluation of a [linear polynomial](#) $a + bX$ at $X = i$ is $a + bi$. However, the evaluation of polynomial $X^2 + 1$ at i is 0, since $i^2 + 1 = 0$. This polynomial is [irreducible](#), i.e., cannot be written as a product of two linear polynomials. Basic facts of [abstract algebra](#) then imply that the [kernel](#) of the above map is an [ideal](#) generated by this polynomial, and that the quotient by this ideal is a field, and that there is an [isomorphism](#)

$$\mathbb{R}[X]/(X^2 + 1) \xrightarrow{\cong} \mathbb{C}$$

between the quotient ring and $\mathbb{C}$. Some authors take this as the definition of $\mathbb{C}$.^{[66][67]} This definition expresses $\mathbb{C}$ as a [quadratic algebra](#).

The field $\mathbb{C}$ is [algebraically closed](#) by the [fundamental theorem of algebra](#), and is therefore the [algebraic closure](#) of $\mathbb{R}$.

Matrix representation of complex numbers

Complex numbers $a + bi$ can also be represented by 2×2 [matrices](#) that have the form^[68]

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Here the entries a and b are real numbers. As the sum and product of two such matrices is again of this form, these matrices form a [subring](#) of the ring of 2×2 matrices.

A simple computation shows that the map

$$a + ib \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is a [ring isomorphism](#) from the field of complex numbers to the ring of these matrices, proving that these matrices form a field. This isomorphism associates the square of the absolute value of a complex number with the [determinant](#) of the corresponding matrix, and the conjugate of a complex number with the [transpose](#) of the matrix.



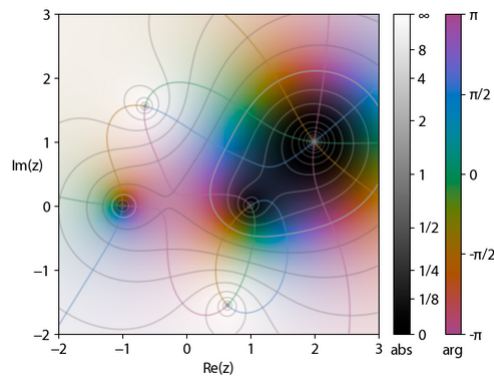
The [polar form](#) representation of complex numbers explicitly gives these matrices as scaled [rotation matrices](#).

$$r(\cos \theta + i \sin \theta) \mapsto \begin{pmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{pmatrix}$$

In particular, the case of $r = 1$, which is $|a + ib| = \sqrt{a^2 + b^2} = 1$, gives (unscaled) rotation matrices.

Комплексный анализ

The study of functions of a complex variable is known as [complex analysis](#) and has enormous practical use in [applied mathematics](#) as well as in other branches of mathematics. Often, the most natural proofs for statements in [real analysis](#) or even [number theory](#) employ techniques from complex analysis (see [prime number theorem](#) for an example).



A [domain coloring](#) graph of the function $\frac{(z^2 - 1)(z - 2 - i)^2}{z^2 + 2 + 2i}$. Darker spots mark moduli near zero, brighter spots are farther away from the origin. The color encodes the argument. The function has zeros for ± 1 , $(2 + i)$ and [poles](#) at $\pm\sqrt{-2 - 2i}$.

Unlike real functions, which are commonly represented as two-dimensional graphs, [complex functions](#) have four-dimensional graphs and may usefully be illustrated by color-coding a [three-dimensional graph](#) to suggest four dimensions, or by animating the complex function's dynamic transformation of the complex plane.



Convergence

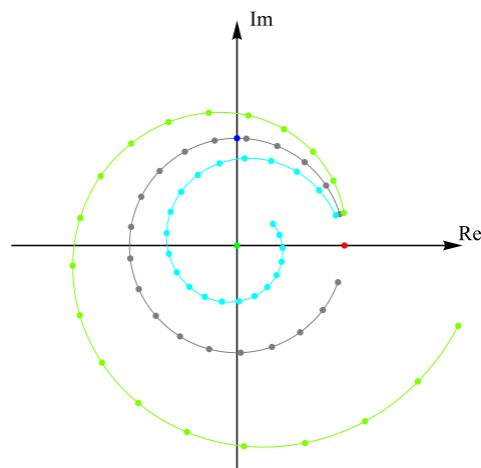


Illustration of the behavior of the sequence z^n for three different values of z (all having the same argument): for $|z| < 1$ the sequence converges to 0 (inner spiral), while it diverges for $|z| > 1$ (outer spiral).

The notions of [convergent series](#) and [continuous functions](#) in (real) analysis have natural analogs in complex analysis. A sequence of complex numbers is said to [converge](#) if and only if its real and imaginary parts do. This is equivalent to the [\(\$\epsilon\$, \$\delta\$ \)-definition of limits](#), where the absolute value of real numbers is replaced by the one of complex numbers. From a more abstract point of view, $\mathbb{C}$, endowed with the [metric](#)

$$d(z_1, z_2) = |z_1 - z_2|$$

is a complete [metric space](#), which notably includes the [triangle inequality](#)

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

for any two complex numbers z_1 and z_2 .

Complex exponential

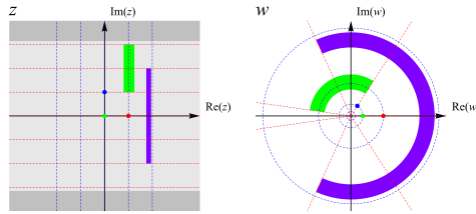


Illustration of the complex exponential function mapping the complex plane, $w = \exp(z)$. The left plane shows a square mesh with mesh size 1, with the three complex numbers 0, 1, and i highlighted. The two rectangles (in magenta and green) are mapped to circular segments, while the lines parallel to the x-axis are mapped to rays emanating from, but not containing the origin. Lines parallel to the y-axis are mapped to circles.

Like in real analysis, this notion of convergence is used to construct a number of [elementary functions](#): the [exponential function](#) $\exp z$, also written e^z , is defined as the [infinite series](#), which can be shown to [converge](#) for any z :

$$\exp z := 1 + z + \frac{z^2}{2 \cdot 1} + \frac{z^3}{3 \cdot 2 \cdot 1} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

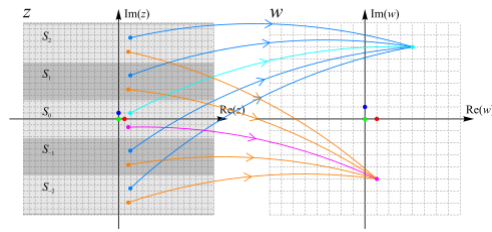
For example, $\exp(1)$ is [Euler's number](#) $e \approx 2.718$. [Euler's formula](#) states:

$$\exp(i\varphi) = \cos \varphi + i \sin \varphi$$

for any real number φ . This formula is a quick consequence of general basic facts about convergent power series and the definitions of the involved functions as power series. As a special case, this includes [Euler's identity](#)

$$\exp(i\pi) = -1.$$

Complex logarithm



The exponential function maps complex numbers z differing by a multiple of $2\pi i$ to the same complex number w .

For any positive real number t , there is a unique real number x such that $\exp(x) = t$. This leads to the definition of the **natural logarithm** as the **inverse** $\ln: \mathbb{R}^+ \rightarrow \mathbb{R}; x \mapsto \ln x$ of the exponential function. The situation is different for complex numbers, since

$$\exp(z + 2\pi i) = \exp z \exp(2\pi i) = \exp z$$

by the functional equation and Euler's identity. For example, $e^{i\pi} = e^{3i\pi} = -1$, so both $i\pi$ and $3i\pi$ are possible values for the complex logarithm of -1 .



In general, given any non-zero complex number w , any number z solving the equation

$$\exp z = w$$

is called a **complex logarithm** of w , denoted $\log w$. It can be shown that these numbers satisfy^[69]

$$z = \log w = \ln |w| + i \arg w,$$

where **arg** is the **argument** defined [above](#), and **ln** the (real) **natural logarithm**. As **arg** is a **multivalued function**, unique only up to a multiple of 2π , **log** is also multivalued. The **principal value** of **log** is often taken by restricting the imaginary part to the **interval** $(-\pi, \pi]$. This leads to the complex logarithm being a **bijective** function taking values in the strip $\mathbb{R}^+ + i(-\pi, \pi]$ (that is denoted S_0 in the above illustration)

$$\ln: \mathbb{C}^\times \rightarrow \mathbb{R}^+ + i(-\pi, \pi].$$

If $z \in \mathbb{C} \setminus (-\mathbb{R}_{\geq 0})$ is not a non-positive real number (a positive or a non-real number), the resulting **principal value** of the complex logarithm is obtained with $-\pi < \varphi < \pi$. It is an **analytic function** outside the negative real numbers, but it cannot be prolonged to a function that is continuous at any negative real number $z \in -\mathbb{R}^+$, where the principal value is $\ln z = \ln(-z) + i\pi$.^[h]

Complex **exponentiation** z^ω is defined as

$$z^\omega = \exp(\omega \ln z),$$

and is multi-valued, except when ω is an integer. For $\omega = 1/n$, for some natural number n , this recovers the non-uniqueness of n th roots mentioned above. If $z > 0$ is real (and ω an arbitrary

complex number), one has a preferred choice of **$\ln x$** , the real logarithm, which can be used to define a preferred exponential function.

Complex numbers, unlike real numbers, do not in general satisfy the unmodified power and logarithm identities, particularly when naïvely treated as single-valued functions; see [failure of power and logarithm identities](#). For example, they do not satisfy

$$a^{bc} = (a^b)^c.$$

Both sides of the equation are multivalued by the definition of complex exponentiation given here, and the values on the left are a subset of those on the right.

Complex sine and cosine

The series defining the real trigonometric functions [sin](#) and [cos](#), as well as the [hyperbolic functions](#) [sinh](#) and [cosh](#), also carry over to complex arguments without change. For the other trigonometric and hyperbolic functions, such as [tan](#), things are slightly more complicated, as the defining series do not converge for all complex values. Therefore, one must define them either in terms of sine, cosine and exponential, or, equivalently, by using the method of [analytic continuation](#).

The value of a trigonometric or hyperbolic function of a complex number can be expressed in terms of those functions evaluated on real numbers, via angle-addition formulas. For $z = x + iy$,

$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

$$\cos z = \cos x \cosh y - i \sin x \sinh y$$

$$\tan z = \frac{\tan x + i \tanh y}{1 - i \tan x \tanh y}$$

$$\cot z = -\frac{1 + i \cot x \coth y}{\cot x - i \coth y}$$

$$\sinh z = \sinh x \cos y + i \cosh x \sin y$$

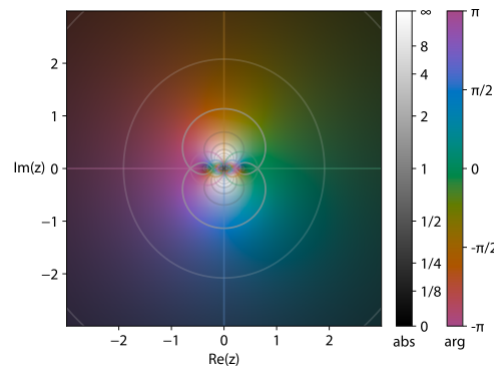
$$\cosh z = \cosh x \cos y + i \sinh x \sin y$$

$$\tanh z = \frac{\tanh x + i \tan y}{1 + i \tanh x \tan y}$$

$$\coth z = \frac{1 - i \coth x \cot y}{\coth x - i \cot y}$$

Where these expressions are not well defined, because a trigonometric or hyperbolic function evaluates to infinity or there is division by zero, they are nonetheless correct as [limits](#).

Holomorphic functions



Color wheel graph of the function $\sin(1/z)$ that is holomorphic except at $z = 0$, which is an essential singularity of this function. White parts inside refer to numbers having large absolute values.

A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is called [holomorphic](#) or *complex differentiable* at a point z_0 if the limit

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists (in which case it is denoted by $f'(z_0)$). This mimics the definition for real differentiable functions, except that all quantities are complex numbers. Loosely speaking, the freedom of approaching z_0 in different directions imposes a much stronger condition than being (real) differentiable. For example, the function

$$f(z) = \bar{z}$$

is differentiable as a function $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, but is *not* complex differentiable. A real differentiable function is complex differentiable if and only if it satisfies the [Cauchy–Riemann equations](#), which are sometimes abbreviated as

$$\frac{\partial f}{\partial \bar{z}} = 0.$$

Complex analysis shows some features not apparent in real analysis. For example, the [identity theorem](#) asserts that two holomorphic functions f and g agree if they agree on an arbitrarily small [open subset](#) of $\mathbb{C}$. [Meromorphic functions](#), functions that can locally be written as $f(z)/(z - z_0)^n$ with a holomorphic function f , still share some of the features of holomorphic functions. Other functions have [essential singularities](#), such as $\sin(1/z)$ at $z = 0$.

Приложения

Complex numbers have applications in many scientific areas, including [signal processing](#), [control theory](#), [electromagnetism](#), [fluid dynamics](#), [quantum mechanics](#), [cartography](#), and [vibration analysis](#). Some of these applications are described below.

Complex conjugation is also employed in [inversive geometry](#), a branch of geometry studying reflections more general than ones about a line. In the [network analysis of electrical circuits](#), the complex conjugate is used in finding the equivalent impedance when the [maximum power transfer theorem](#) is looked for.

Geometry

Shapes

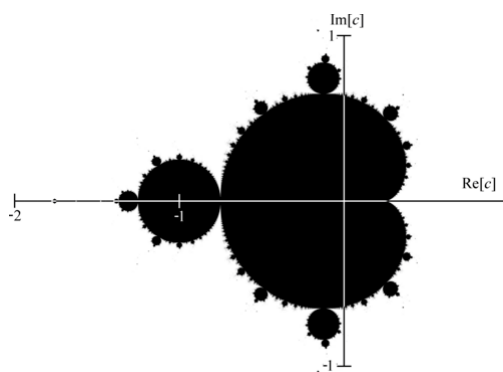
Three [non-collinear](#) points u, v, w in the plane determine the [shape](#) of the triangle $\{u, v, w\}$.

Locating the points in the complex plane, this shape of a triangle may be expressed by complex arithmetic as

$$S(u, v, w) = \frac{u - w}{u - v}.$$

The shape S of a triangle will remain the same, when the complex plane is transformed by translation or dilation (by an [affine transformation](#)), corresponding to the intuitive notion of shape, and describing [similarity](#). Thus each triangle $\{u, v, w\}$ is in a [similarity class](#) of triangles with the same shape.^[70]

Fractal geometry



The Mandelbrot set with the real and imaginary axes labeled.

The [Mandelbrot set](#) is a popular example of a fractal formed on the complex plane. It is defined by plotting every location c where iterating the sequence $f_c(z) = z^2 + c$ does not [diverge](#) when [iterated](#) infinitely. Similarly, [Julia sets](#) have the same rules, except where c remains constant.

Triangles

Every triangle has a unique [Steiner inellipse](#) – an [ellipse](#) inside the triangle and tangent to the midpoints of the three sides of the triangle. The [foci](#) of a triangle's Steiner inellipse can be found as follows, according to [Marden's theorem](#).^{[71][72]} Denote the triangle's vertices in the complex plane as $a = x_A + y_A i$, $b = x_B + y_B i$, and $c = x_C + y_C i$. Write the [cubic equation](#) $(x - a)(x - b)(x - c) = 0$, take its derivative, and equate the (quadratic) derivative to zero. Marden's theorem says that the solutions of this equation are the complex numbers denoting the locations of the two foci of the Steiner inellipse.

Algebraic number theory

•



Construction of a regular
pentagon [using straightedge
and compass](#).

As mentioned above, any nonconstant polynomial equation (in complex coefficients) has a solution in $\mathbb{C}$. *A fortiori*, the same is true if the equation has rational coefficients. The roots of such equations are called [algebraic numbers](#) – they are a principal object of study in [algebraic number theory](#). Compared to $\overline{\mathbb{Q}}$, the algebraic closure of $\mathbb{Q}$, which also contains all algebraic numbers, $\mathbb{C}$ has the advantage of being easily understandable in geometric terms. In this way, algebraic methods can be used to study geometric questions and vice versa. With algebraic methods, more specifically applying the machinery of [field theory](#) to the [number field](#) containing [roots of unity](#), it can be shown that it is not possible to construct a regular [nonagon using only compass and straightedge](#) – a purely geometric problem.

Another example is the [Gaussian integers](#); that is, numbers of the form $x + iy$, where x and y are integers, which can be used to classify [sums of squares](#).

Analytic number theory

Analytic number theory studies numbers, often integers or rationals, by taking advantage of the fact that they can be regarded as complex numbers, in which analytic methods can be used. This is done by encoding number-theoretic information in complex-valued functions. For example, the [Riemann zeta function](#) $\zeta(s)$ is related to the distribution of [prime numbers](#).

Improper integrals

In applied fields, complex numbers are often used to compute certain real-valued [improper integrals](#), by means of complex-valued functions. Several methods exist to do this; see [methods of contour integration](#).

Dynamic equations

In [differential equations](#), it is common to first find all complex roots r of the [characteristic equation](#) of a [linear differential equation](#) or equation system and then attempt to solve the system in terms of base functions of the form $f(t) = e^{rt}$. Likewise, in [difference equations](#), the complex roots r of the characteristic equation of the difference equation system are used, to attempt to solve the system in terms of base functions of the form $f(t) = r^t$.

Linear algebra

Since $\mathbb{C}$ is algebraically closed, any non-empty complex [square matrix](#) has at least one (complex) [eigenvalue](#). By comparison, real matrices do not always have real eigenvalues, for example [rotation matrices](#) (for rotations of the plane for angles other than 0° or 180°) leave no direction fixed, and therefore do not have any *real* eigenvalue. The existence of (complex) eigenvalues, and the ensuing existence of [eigendecomposition](#) is a useful tool for computing matrix powers and [matrix exponentials](#).



Complex numbers often generalize concepts originally conceived in the real numbers. For example, the [conjugate transpose](#) generalizes the [transpose](#), [hermitian matrices](#) generalize [symmetric matrices](#), and [unitary matrices](#) generalize [orthogonal matrices](#).

In applied mathematics

Control theory

In [control theory](#), systems are often transformed from the [time domain](#) to the complex [frequency domain](#) using the [Laplace transform](#). The system's [zeros and poles](#) are then analyzed in the *complex plane*. The [root locus](#), [Nyquist plot](#), and [Nichols plot](#) techniques all make use of the complex plane.

In the root locus method, it is important whether zeros and poles are in the left or right half planes, that is, have real part greater than or less than zero. If a linear, time-invariant (LTI) system has poles that are

- in the right half plane, it will be [unstable](#),
- all in the left half plane, it will be [stable](#),
- on the imaginary axis, it will have [marginal stability](#).

If a system has zeros in the right half plane, it is a [nonminimum phase](#) system.

Signal analysis

Complex numbers are used in [signal analysis](#) and other fields for a convenient description for periodically varying signals. For given real functions representing actual physical quantities, often in terms of sines and cosines, corresponding complex functions are considered of which the real parts are the original quantities. For a [sine wave](#) of a given [frequency](#), the absolute value $|Z|$ of the corresponding Z is the [amplitude](#) and the [argument](#) $\arg Z$ is the [phase](#).

If [Fourier analysis](#) is employed to write a given real-valued signal as a sum of periodic functions, these periodic functions are often written as complex-valued functions of the form

$$x(t) = \operatorname{Re}\{X(t)\}$$

and

$$X(t) = Ae^{i\omega t} = ae^{i\phi}e^{i\omega t} = ae^{i(\omega t + \phi)}$$



where ω represents the [angular frequency](#) and the complex number A encodes the phase and amplitude as explained above.

This use is also extended into [digital signal processing](#) and [digital image processing](#), which use digital versions of Fourier analysis (and [wavelet](#) analysis) to transmit, [compress](#), restore, and otherwise process [digital audio](#) signals, still images, and [video](#) signals.

Another example, relevant to the two side bands of [amplitude modulation](#) of AM radio, is:

$$\begin{aligned}\cos((\omega + \alpha)t) + \cos((\omega - \alpha)t) &= \operatorname{Re}\left(e^{i(\omega + \alpha)t} + e^{i(\omega - \alpha)t}\right) \\ &= \operatorname{Re}\left((e^{i\alpha t} + e^{-i\alpha t}) \cdot e^{i\omega t}\right) \\ &= \operatorname{Re}\left(2 \cos(\alpha t) \cdot e^{i\omega t}\right) \\ &= 2 \cos(\alpha t) \cdot \operatorname{Re}\left(e^{i\omega t}\right) \\ &= 2 \cos(\alpha t) \cdot \cos(\omega t).\end{aligned}$$

In physics

Electromagnetism and electrical engineering

In [electrical engineering](#), the [Fourier transform](#) is used to analyze varying [electric currents](#) and [voltages](#). The treatment of [resistors](#), [capacitors](#), and [inductors](#) can then be unified by introducing imaginary, frequency-dependent resistances for the latter two and combining all three in a single complex number called the [impedance](#). This approach is called [phasor](#) calculus.

In electrical engineering, the imaginary unit is denoted by j , to avoid confusion with I , which is generally in use to denote electric current, or, more particularly, i , which is generally in use to

denote instantaneous electric current.

Because the voltage in an AC circuit is oscillating, it can be represented as

$$V(t) = V_0 e^{j\omega t} = V_0 (\cos \omega t + j \sin \omega t),$$

To obtain the measurable quantity, the real part is taken:

$$v(t) = \operatorname{Re}(V) = \operatorname{Re}[V_0 e^{j\omega t}] = V_0 \cos \omega t.$$

The complex-valued signal $V(t)$ is called the [analytic](#) representation of the real-valued, measurable signal $v(t)$. ^[73]

Fluid dynamics

In [fluid dynamics](#), complex functions are used to describe [potential flow in two dimensions](#).



Quantum mechanics

The complex number field is intrinsic to the [mathematical formulations of quantum mechanics](#), where complex [Hilbert spaces](#) provide the context for one such formulation that is convenient and perhaps most standard. The original foundation formulas of quantum mechanics – the [Schrödinger equation](#) and Heisenberg's [matrix mechanics](#) – make use of complex numbers. ^[74]

Relativity

In [special relativity](#) and [general relativity](#), some formulas for the metric on [spacetime](#) become simpler if one takes the time component of the spacetime continuum to be imaginary. (This approach is no longer standard in classical relativity, but is [used in an essential way](#) in [quantum field theory](#).) Complex numbers are essential to [spinors](#), which are a generalization of the [tensors](#) used in relativity. ^[75]

Характеристики, обобщения и связанные с ними понятия

Algebraic characterization

The field $\mathbb{C}$ has the following three properties:

- First, it has [characteristic](#) 0. This means that $1 + 1 + \dots + 1 \neq 0$ for any number of summands (all of which equal one).
- Second, its [transcendence degree](#) over $\mathbb{Q}$, the [prime field](#) of $\mathbb{C}$, is the [cardinality of the continuum](#).

- Third, it is [algebraically closed](#) (see above).

It can be shown that any field having these properties is [isomorphic](#) (as a field) to $\mathbb{C}$. For example, the [algebraic closure](#) of the field $\mathbb{Q}_p$ of the [p-adic number](#) also satisfies these three properties, so these two fields are isomorphic (as fields, but not as topological fields).^[76] Also, $\mathbb{C}$ is isomorphic to the field of complex [Puiseux series](#). However, specifying an isomorphism requires the [axiom of choice](#). Another consequence of this algebraic characterization is that $\mathbb{C}$ contains many proper subfields that are isomorphic to $\mathbb{C}$.

Characterization as a topological field

The preceding characterization of $\mathbb{C}$ describes only the algebraic aspects of $\mathbb{C}$. That is to say, the properties of [nearness](#) and [continuity](#), which matter in areas such as [analysis](#) and [topology](#), are not dealt with. The following description of $\mathbb{C}$ as a [topological field](#) (that is, a field that is equipped with a [topology](#), which allows the notion of convergence) does take into account the topological properties. $\mathbb{C}$ contains a subset P (namely the set of positive real numbers) of nonzero elements satisfying the following three conditions:

- P is closed under addition, multiplication and taking inverses.
- If x and y are distinct elements of P , then either $x - y$ or $y - x$ is in P .
- If S is any nonempty subset of P , then $S + P = x + P$ for some x in $\mathbb{C}$.

Moreover, $\mathbb{C}$ has a nontrivial [involutive automorphism](#) $x \mapsto x^*$ (namely complex conjugation), such that $x x^*$ is in P for any nonzero x in $\mathbb{C}$.

Any field F with these properties can be endowed with a topology by taking the sets $B(x, p) = \{y \mid p - (y - x)(y - x)^* \in P\}$ as a [base](#), where x ranges over the field and p ranges over P . With this topology F is isomorphic as a *topological field* to $\mathbb{C}$.

The only [connected locally compact topological fields](#) are $\mathbb{R}$ and $\mathbb{C}$. This gives another characterization of $\mathbb{C}$ as a topological field, because $\mathbb{C}$ can be distinguished from $\mathbb{R}$ because the nonzero complex numbers are [connected](#), while the nonzero real numbers are not.^[77]

Other number systems



Number systems

	rational numbers $\mathbb{Q}$	real numbers $\mathbb{R}$	complex numbers $\mathbb{C}$	quaternions $\mathbb{H}$	octonions $\mathbb{O}$	sedenions $\mathbb{S}$
complete	No	Yes	Yes	Yes	Yes	Yes
dimension as an $\mathbb{R}$-vector space	[does not apply]	1	2	4	8	16
ordered	Yes	Yes	No	No	No	No
multiplication commutative ($xy = yx$)	Yes	Yes	Yes	No	No	No
multiplication associative ($(xy)z = x(yz)$)	Yes	Yes	Yes	Yes	No	No
normed division algebra (over $\mathbb{R}$)	[does not apply]	Yes	Yes	Yes	Yes	No



The process of extending the field $\mathbb{R}$ of reals to $\mathbb{C}$ is an instance of the *Cayley–Dickson construction*. Applying this construction iteratively to $\mathbb{C}$ then yields the **quaternions**, the **octonions**,^[78] the **sedenions**, and the **trigintaduonions**. This construction turns out to diminish the structural properties of the involved number systems.

Unlike the reals, $\mathbb{C}$ is not an **ordered field**, that is to say, it is not possible to define a relation $z_1 < z_2$ that is compatible with the addition and multiplication. In fact, in any ordered field, the square of any element is necessarily positive, so $i^2 = -1$ precludes the existence of an **ordering** on $\mathbb{C}$.^[79] Passing from $\mathbb{C}$ to the quaternions $\mathbb{H}$ loses commutativity, while the octonions (additionally to not being commutative) fail to be associative. The reals, complex numbers, quaternions and octonions are all **normed division algebras** over $\mathbb{R}$. By **Hurwitz's theorem** they are the only ones; the **sedenions**, the next step in the Cayley–Dickson construction, fail to have this structure.

The Cayley–Dickson construction is closely related to the **regular representation** of $\mathbb{C}$, thought of as an **$\mathbb{R}$ -algebra** (an $\mathbb{R}$ -vector space with a multiplication), with respect to the basis $(1, i)$. This means the following: the $\mathbb{R}$ -linear map

$$\begin{aligned}\mathbb{C} &\rightarrow \mathbb{C} \\ z &\mapsto wz\end{aligned}$$

for some fixed complex number w can be represented by a 2×2 matrix (once a basis has been chosen). With respect to the basis $(1, i)$, this matrix is

$$\begin{pmatrix} \operatorname{Re}(w) & -\operatorname{Im}(w) \\ \operatorname{Im}(w) & \operatorname{Re}(w) \end{pmatrix},$$

that is, the one mentioned in the section on matrix representation of complex numbers above. While this is a **linear representation** of $\mathbb{C}$ in the 2×2 real matrices, it is not the only one. Any

matrix

$$J = \begin{pmatrix} p & q \\ r & -p \end{pmatrix}, \quad p^2 + qr + 1 = 0$$

has the property that its square is the negative of the identity matrix: $J^2 = -I$. Then

$$\{z = aI + bJ : a, b \in \mathbb{R}\}$$

is also isomorphic to the field $\mathbb{C}$, and gives an alternative complex structure on $\mathbb{R}^2$. This is generalized by the notion of a [linear complex structure](#).

[Hypercomplex numbers](#) also generalize $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$. For example, this notion contains the [split-complex numbers](#), which are elements of the ring $\mathbb{R}[x]/(x^2 - 1)$ (as opposed to $\mathbb{R}[x]/(x^2 + 1)$ for complex numbers). In this ring, the equation $a^2 = 1$ has four solutions.

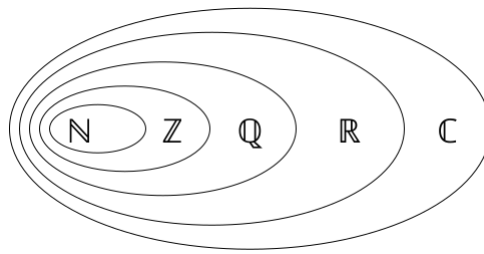
The field $\mathbb{R}$ is the completion of $\mathbb{Q}$, the field of [rational numbers](#), with respect to the usual [absolute value metric](#). Other choices of [metrics](#) on $\mathbb{Q}$ lead to the fields $\mathbb{Q}_p$ of [p-adic numbers](#) (for any [prime number](#) p), which are thereby analogous to $\mathbb{R}$. There are no other nontrivial ways of completing $\mathbb{Q}$ than $\mathbb{R}$ and $\mathbb{Q}_p$, by [Ostrowski's theorem](#). The algebraic closures $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ still carry a norm, but (unlike $\mathbb{C}$) are not complete with respect to it. The completion $\mathbb{C}_p$ of $\overline{\mathbb{Q}_p}$ turns out to be algebraically closed. By analogy, $\mathbb{C}_p$ is called the p -adic complex numbers.

The fields $\mathbb{R}$, $\mathbb{Q}_p$, and their finite field extensions, including $\mathbb{C}$, are called [local fields](#).



Смотрите также

- [Analytic continuation](#)
- [Circular motion using complex numbers](#)
- [Complex-base system](#)
- [Complex coordinate space](#)
- [Complex geometry](#)
- [Geometry of numbers](#)
- [Dual-complex number](#)
- [Eisenstein integer](#)
- [Geometric algebra](#) (which includes the complex plane as the 2-dimensional [spinor](#) subspace $\mathcal{G}_2^+$)



Set inclusions between the [natural numbers](#) ($\mathbb{N}$), the [integers](#) ($\mathbb{Z}$), the [rational numbers](#) ($\mathbb{Q}$), the [real numbers](#) ($\mathbb{R}$), and the [complex numbers](#) ($\mathbb{C}$)

Примечания

- a. [Solomentsev 2001](#): "The plane $\mathbb{R}^2$ whose points are identified with the elements of $\mathbb{C}$ is called the complex plane ... The complete geometric interpretation of complex numbers and operations on them appeared first in the work of C. Wessel (1799). The geometric representation of complex numbers, sometimes called the 'Argand diagram', came into use after the publication in 1806 and 1814 of papers by J.R. Argand, who rediscovered, largely independently, the findings of Wessel". 
- b. In the literature the imaginary unit often precedes the radical sign, even when preceded itself by an integer. ^[32]
- c. It has been proved that imaginary numbers necessarily appear in the cubic formula when the equation has three real, different roots by Pierre Laurent Wantzel in 1843, Vincenzo Mollame in 1890, Otto Hölder in 1891, and Adolf Kneser in 1892. Paolo Ruffini also provided an incomplete proof in 1799.—S. Confalonieri (2015) ^[34]
- d. [Argand 1814](#), p. 204 defines the modulus of a complex number but he doesn't name it: *"Dans ce qui suit, les accents, indifféremment placés, seront employés pour indiquer la grandeur absolue des quantités qu'ils affectent; ainsi, si $a = m + n\sqrt{-1}$, m et n étant réels, on devra entendre que a ou $a' = \sqrt{m^2 + n^2}$."*
[In what follows, accent marks, wherever they're placed, will be used to indicate the absolute size of the quantities to which they're assigned; thus if $a = m + n\sqrt{-1}$, m and n being real, one should understand that a or $a' = \sqrt{m^2 + n^2}$.]
[Argand 1814](#), p. 208 defines and names the *module* and the *direction factor* of a complex number: "... $a = \sqrt{m^2 + n^2}$ pourrait être appelé le module de $a + b\sqrt{-1}$, et représenterait la grandeur absolue de la ligne $a + b\sqrt{-1}$, tandis que l'autre facteur, dont le module est l'unité, en représenterait la direction."
[... $a = \sqrt{m^2 + n^2}$ could be called the *module* of $a + b\sqrt{-1}$ and would represent the *absolute size* of the line $a + b\sqrt{-1}$, (Argand represented complex numbers as vectors.)

whereas the other factor [namely, $\frac{a}{\sqrt{a^2+b^2}} + \frac{b}{\sqrt{a^2+b^2}}\sqrt{-1}$], whose module is unity [1], would represent its direction.]

- e. Gauss writes:^[56] *"Quemadmodum scilicet arithmetica sublimior in quaestionibus hactenus pertractatis inter solos numeros integros reales versatur, ita theoremata circa residua biquadratica tunc tantum in summa simplicitate ac genuina venustate resplendent, quando campus arithmeticae ad quantitates imaginarias extenditur, ita ut absque restrictione ipsius obiectum constituent numeri formae $a + bi$, denotantibus i , pro more quantitatem imaginariam $\sqrt{-1}$, atque a, b indefinite omnes numeros reales integros inter $-\infty$ et $+\infty$."* [Of course just as the higher arithmetic has been investigated so far in problems only among real integer numbers, so theorems regarding biquadratic residues then shine in greatest simplicity and genuine beauty, when the field of arithmetic is extended to *imaginary* quantities, so that, without restrictions on it, numbers of the form $a + bi - i$ denoting by convention the imaginary quantity $\sqrt{-1}$, and the variables a, b [denoting] all real integer numbers between $-\infty$ and $+\infty$ – constitute an object.]
- f. Gauss:^[57] *"Tales numeros vocabimus numeros integros complexos, ita quidem, ut reales complexis non opponantur, sed tamquam species sub his contineri censeantur."* [We will call such numbers [namely, numbers of the form $a + bi$] "complex integer numbers", so that real [numbers] are regarded not as the opposite of complex [numbers] but [as] a type [of number that] is, so to speak, contained within them.]
- g. Gauss:^[58] *"Productum numeri complexi per numerum ipsi conjunctum utriusque normam vocamus. Pro norma itaque numeri realis, ipsius quadratum habendum est."* [We call a "norm" the product of a complex number [for example, $a + ib$] with its conjugate $[a - ib]$. Therefore the square of a real number should be regarded as its norm.]
- h. However for another inverse function of the complex exponential function (and not the above defined principal value), the branch cut could be taken at any other [ray](#) thru the origin.



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